DIGITAL DIRECT MANUFACTURING

CAD Model Dimension vs Printed Dimension Accuracy

Material: PLA | Process: FFF / FDM | March 2026

1. Study Overview

This report presents the findings of a dimensional accuracy study conducted as part of the Digital Direct Manufacturing (DDM) course. The objective was to compare CAD model dimensions with actual printed dimensions using PLA filament on an FFF/FDM 3D printer, and to identify causes of deviation and appropriate corrective measures.

2. Summary Statistics

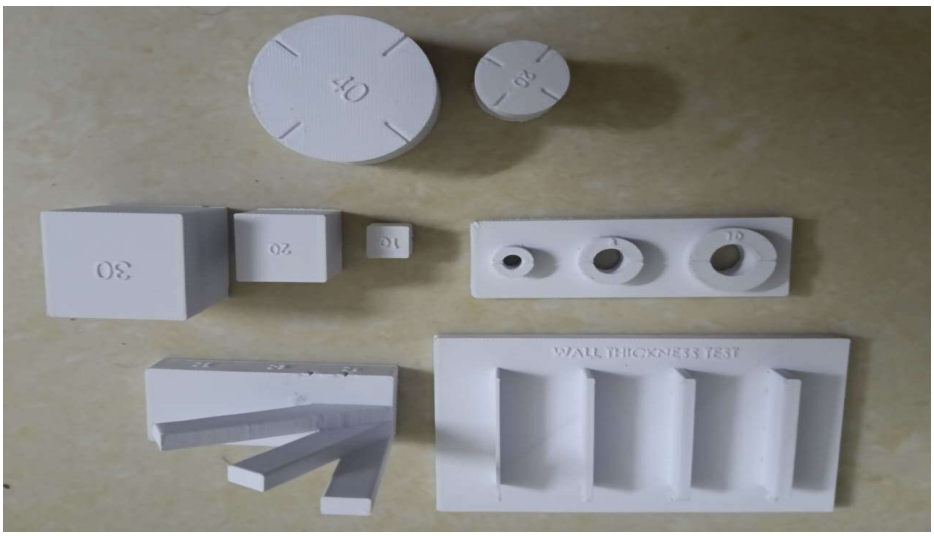
Total Measurements	Shapes Tested	Avg |Error|	Max Deviation
16	4	0.172 mm	0.450 mm
Dimensional Accuracy	**Material**		
99.1%	PLA		

3. Measurement Data — All Geometries

Error = Printed − CAD. Negative values indicate under-size; positive values indicate over-size.

Shape	Feature	CAD (mm)	Printed (mm)	Error (mm)	Error (%)	Speed (mm/s)	Defects
Calibration Cube	**X Axis**	20.00	19.82	**-0.180**	-0.90%	60	None
Calibration Cube	**Y Axis**	20.00	19.87	**-0.130**	-0.65%	60	None
Calibration Cube	**Z Axis**	20.00	19.94	**-0.060**	-0.30%	60	None
Calibration Cube	**X Axis**	20.00	19.71	**-0.290**	-1.45%	80	Ghosting/Ringing
Calibration Cube	**Y Axis**	20.00	19.68	**-0.320**	-1.60%	80	Ghosting/Ringing
Calibration Cube	**X Axis**	20.00	19.91	**-0.090**	-0.45%	40	None
Cylinder	**Diameter**	20.00	19.74	**-0.260**	-1.30%	60	None
Cylinder	**Height**	20.00	19.96	**-0.040**	-0.20%	60	None
Cylinder	**Diameter**	20.00	19.55	**-0.450**	-2.25%	80	Over-extrusion
Hole Test Plate	**Hole Ø5mm**	5.00	4.71	**-0.290**	-5.80%	60	Hole Shrinkage
Hole Test Plate	**Hole Ø10mm**	10.00	9.78	**-0.220**	-2.20%	60	Hole Shrinkage
Hole Test Plate	**Hole Ø15mm**	15.00	14.81	**-0.190**	-1.27%	60	Hole Shrinkage
Wall Thickness	**0.5mm Wall**	0.50	0.62	**+0.120**	+24.00%	60	Over-extrusion
Wall Thickness	**1.0mm Wall**	1.00	1.08	**+0.080**	+8.00%	60	None

Shape	Feature	CAD (mm)	Printed (mm)	Error (mm)	Error (%)	Speed (mm/s)	Defects
Wall Thickness	**2.0mm Wall**	2.00	2.03	**+0.030**	+1.50%	60	None
Wall Thickness	**3.0mm Wall**	3.00	2.99	**-0.010**	-0.33%	60	None



(Total 3D printed parts for assessment)



(Calibration of Cube)



(Calibration of Wall thickness)



(Calibration of Cylinder)


(Hole test Plate)


(Calibration of Overhang)

4. Print Speed Effect on Dimensional Accuracy

The same calibration cube geometry was printed at three speeds (40, 60, 80 mm/s) with all other parameters held constant (Layer: 0.2 mm, Infill: 20%, Pattern: Grid).

Print Speed (mm/s)	Avg |Error| (mm)	Observation
40 mm/s (Slow)	**0.090 mm**	Best accuracy — slow, consistent extrusion
60 mm/s (Standard)	**0.134 mm**	Good balance of accuracy and print time
80 mm/s (Fast)	**0.353 mm**	Highest error — vibration & ringing artefacts

5. Error by Shape — Visual Summary

Average absolute dimensional error grouped by test geometry type.

Shape	Avg |Error| — Visual	mm
Calibration Cube		0.178
Cylinder		0.250
Hole Test Plate		0.233
Wall Thickness		0.060

6. Defect Analysis & Corrective Solutions

The following print defects were observed across test specimens and their impact on dimensional accuracy is documented below.

Defect	Frequency	Cause	Recommended Solution
Hole Shrinkage	High	PLA contracts inward on concave geometry	Apply +0.1 to +0.2 mm horizontal expansion offset in slicer
Ghosting / Ringing	Medium	Vibration at high speed (80 mm/s) near corners	Reduce speed to ≤60 mm/s; check belt tension
Over-extrusion	Low	Flow rate above 100%; thin walls worse affected	Calibrate e-steps; set flow to 95–98%; measure filament Ø
Elephant Foot	Low	Excessive first-layer squish on heated bed	Increase Z offset; lower bed temp to 55°C for PLA

7. Conclusion

In this study, **16 measurements** were recorded across **4 test geometry types** using PLA material. The average absolute dimensional deviation was **0.172 mm**, with a maximum deviation of **0.450 mm**. The overall dimensional accuracy achieved was **99.1%**.

Lower print speed (40 mm/s) consistently produced the smallest dimensional error. Hole features showed the greatest under-size deviation due to inward thermal contraction of PLA, a well-known characteristic of FFF printing. Wall thickness tests confirmed that features below 1 mm are susceptible to over-extrusion artefacts.

Key Recommendations: (1) Calibrate e-steps and flow rate for PLA. (2) Apply horizontal expansion offset (+0.10 to +0.15 mm) for hole features. (3) Use 40–60 mm/s for dimensional-critical parts. (4) Employ 0.1–0.2 mm layer height for improved Z accuracy.